The Effect of Group Therapy on Depression in Rural Ghana: Evidence from a Randomized Controlled Trial

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Abstract

This paper evaluates the effectiveness of a group therapy (GT) intervention on reducing depression among individuals in rural Ghana using data from a cluster-randomized controlled trial. Employing the Kessler-10 (K10) psychological distress scale as the primary outcome measure, I find that GT sessions reduced depression scores by 0.54 points (0.1 standard deviations) on average ($p = 0.001$). However, average effects mask substantial heterogeneity by gender. While women in treated households experienced a significant reduction of 0.85 points ($p < 0.001$), men showed no statistically significant improvement. Exploratory analysis of baseline characteristics reveals that women report higher initial depression levels than men, while household wealth shares no significant association with mental health outcomes. These findings suggest that group-based mental health interventions may be particularly effective for women in low-income settings, offering critical implications for the targeting of mental health policy in developing countries.

1 Introduction

Mental health disorders represent a significant yet frequently overlooked burden in low- and middle-income countries (LMICs). Depression, in particular, affects hundreds of millions globally and is associated with reduced labor productivity, impaired social functioning, and diminished quality of life (WHO, 2017). Despite the magnitude of this challenge, mental health services remain severely underprovided in developing countries, where fewer than 10% of those in need receive adequate treatment (Patel et al., 2007).

In recent years, researchers and policymakers have increasingly turned to scalable, community-based interventions to address this treatment gap. Group therapy (GT) programs, which leverage peer support and shared experiences, offer a potentially cost-effective approach to delivering mental health services in resource-constrained settings. However, rigorous evidence regarding the effectiveness of such interventions in Sub-Saharan Africa remains limited.

This paper contributes to the growing literature on mental health interventions in development economics by evaluating a randomized controlled trial (RCT) of group therapy sessions conducted in rural Ghana. The intervention targeted households across multiple villages, with treatment randomly assigned at the household level. Using the Kessler-10 (K10) psychological distress scale as the primary outcome, I assess both the aggregate intent-to-treat (ITT) effect and heterogeneity in treatment effects by gender.

The analysis yields three primary findings. First, GT sessions were effective at reducing depression on average: treated individuals scored 0.54 points lower on the K10 scale compared to the control group ($p = 0.001$). Second, treatment effects differed markedly by gender. While women in treated households experienced a significant 0.85-point reduction in depression scores ($p < 0.001$), men showed no statistically significant improvement. Third, baseline exploratory analysis reveals that women report higher levels of depression than men, whereas household wealth—proxied by total asset values—shows no significant correlation with mental health outcomes.

These findings have salient implications for the design of mental health interventions in low-income settings. The differential treatment effects suggest that group therapy may be particularly well-suited to addressing women's mental health needs, potentially due to differences in social support mechanisms or a greater willingness to engage with therapeutic processes. Conversely, the lack of observable effects for men points to the necessity of alternative or complementary approaches to effectively reach this demographic.

The remainder of this paper is organized as follows. Section 2 describes the data sources and sample construction. Section 3 outlines the empirical methodology. Section 4 presents the main results, including exploratory analysis of baseline characteristics and estimation of treatment effects. Section 5 discusses the implications of these findings and concludes.

2 Data

2.1 Data Sources and Sample Construction

The data for this study are drawn from a household survey conducted across multiple villages in rural Ghana. The survey was administered in two waves: a baseline survey (Wave 1) conducted prior to the intervention and a follow-up survey (Wave 2) conducted six months later, following the completion of the group therapy program. The dataset integrates information from three distinct modules: demographics, assets, and mental health.

The *demographics module* contains individual-level information on household members, including age, gender, and household identifiers. Each observation represents one individual in a given survey wave, uniquely identified by household ID, household member ID, and a wave indicator. This module also records the household-level treatment assignment status.

The *assets module* provides detailed information on household wealth, specifically the quantity and monetary value of assets owned. Assets are categorized into three types: animals (e.g., chickens, goats), tools (e.g., cutlass, hoe), and durable goods (e.g., radio, furniture). For each asset type, the survey records both the quantity owned and the current monetary value per unit.

The *mental health module* contains responses to the Kessler-10 (K10) psychological distress scale, a widely validated instrument for measuring symptoms of depression and anxiety. The K10 consists of 10 questions asking respondents how often they experienced specific symptoms over the past 30 days, with responses ranging from 1 ("none of the time") to 5 ("all of the time"). Total scores range from 10 to 50, with higher scores indicating greater psychological distress.

To construct the analysis sample, I merged the three modules at the individual-wave level. The final dataset contains a maximum of two observations per individual (one per wave). After excluding observations with missing key variables, the final analysis sample comprises 7,398 individuals at baseline (Wave 1) and 6,385 individuals at follow-up (Wave 2).

2.2 Variable Construction

2.2.1 Depression Measures

The primary outcome variable is the Kessler-10 score (*kessler_score*), computed as the sum of responses to all 10 K10 questions. To address item non-response, I imputed missing values for respondents with 1–2 missing items using the mean of available items scaled to the full 10-item total. Respondents with 3 or more missing responses were excluded from the analysis to ensure measurement reliability.

I also constructed a categorical variable (*kessler_categories*) based on standard K10 clinical cut-offs:

- 10–19: No significant depression
- 20–24: Mild depression
- 25–29: Moderate depression
- 30–50: Severe depression

2.2.2 Wealth Measures

To proxy for household wealth, I computed the total monetary value of all assets owned by each household. To address missing valuation data, I imputed missing unit values using the median value of assets within the same specific type (e.g., substituting the median value of "chickens" for missing chicken values). Total asset value was then calculated as the sum of quantity multiplied by the (imputed) unit value across all asset holdings. Given the right-skewed nature of asset distributions, I employed logarithmic transformations for visualization where appropriate.

2.2.3 Demographic Variables

Key demographic variables include age (in years), gender (coded as 1 for female, 0 for male), and household size (proxied by the number of household members surveyed in Wave 1).

3 Empirical Methodology

3.1 Experimental Design

The group therapy intervention followed a cluster-randomized design, with treatment assigned at the household level. Households assigned to the treatment group received access to facilitated group therapy sessions, while control households received no intervention. Because all individuals within a treated household were exposed to the intervention, I cluster standard errors at the household level in all regression specifications to account for intra-household correlation.

3.2 Estimation Strategy

To estimate the average treatment effect (ATE) of GT sessions on depression, I estimate the following linear regression model using Wave 2 data:

$$Y_{ih} = \beta_0 + \beta_1 \cdot \text{Treatment}_h + \varepsilon_{ih} \quad (1)$$

where Y_{ih} is the Kessler-10 score for individual i in household h, Treatment_h is an indicator equal to 1 if the household was assigned to the treatment group, and ε_{ih} is the error term. The coefficient β_1 captures the intent-to-treat (ITT) effect—the difference in mean depression scores between the treatment and control groups.

To examine heterogeneity in treatment effects by gender, I extend the model to include an interaction term:

$$Y_{ih} = \beta_0 + \beta_1 \cdot \text{Treatment}_h + \beta_2 \cdot \text{Woman}_i + \beta_3 \cdot (\text{Treatment}_h \times \text{Woman}_i) + \varepsilon_{ih} \quad (2)$$

where Woman_i is an indicator for female gender. In this specification:

- β_0 represents the mean outcome for untreated men (the reference group).
- β_1 captures the treatment effect for men.
- β_2 represents the gender gap in depression among untreated individuals.
- β_3 captures the differential treatment effect for women relative to men.

The total treatment effect for women is given by the linear combination $\beta_1 + \beta_3$. All regressions employ robust standard errors clustered at the household level.

4 Results

4.1 Descriptive Statistics and Baseline Analysis

Table 1 presents summary statistics for the baseline sample (Wave 1). The mean Kessler-10 score is 20.34, with substantial variation across individuals (SD = 6.31). Scores range from the minimum possible value of 10 to the maximum of 50, indicating that the full spectrum of psychological distress is represented in the sample.

Table 1: Summary Statistics: Baseline Sample (Wave 1)

Variable	N	Mean	SD	Min	Max
Kessler Score	7,398	20.34	6.31	10	50
Total Assets (GHS)	18,778	12,718	105,284	1.5	5,790,900
Female	7,398	0.55	0.50	0	1
Age	7,398	30.10	21.43	0	107

Household wealth, measured by total asset value, exhibits considerable right-skewness, with a mean of 12,718 Ghanaian cedis (GHS) but a maximum exceeding 5.7 million. This distribution reflects the presence of a small number of relatively wealthy households alongside a majority with modest asset holdings.

4.1.1 Depression and Household Wealth

A central question in the mental health literature concerns the gradient between socioeconomic status and psychological well-being. Table 2 reports the correlation between Kessler scores and total household assets, along with regression estimates.

Table 2: Depression and Household Wealth

	Correlation	OLS Coefficient
Total Assets	-0.006	-3.14×10^{-7}
		(0.608)
Observations	7,384	7,384

Note: p-values in parentheses.

Contrary to the hypothesis that poverty exacerbates mental distress, I find no significant relationship between household wealth and depression in this sample. The correlation coefficient is negligible ($r = -0.006$), and the regression coefficient is statistically indistinguishable from zero ($p = 0.608$). This null finding may reflect limited variation in wealth within this predominantly rural, low-income population, or it may suggest that non-material factors—such as social support or physical health—are more salient determinants of mental health in this context.

4.1.2 Depression and Gender

Table 3 examines gender differences in depression at baseline. Women report significantly higher Kessler scores than men, with a mean of 20.82 compared to 19.75 for men.

Table 3: Depression by Gender (Baseline)

Gender	N	Mean	SD	Difference	p-value
Male	3,318	19.75	6.24	—	—
Female	4,080	20.82	6.32	1.07	< 0.001

The gender gap of 1.07 points is statistically significant ($p < 0.001$) and represents approximately 17% of a standard deviation in Kessler scores. This aligns with global evidence documenting higher rates of depression among women, potentially reflecting differential exposure to stressors or gendered social roles.

4.1.3 Depression and Age

I further examine the relationship between age and depression. Table 4 reports the correlation and regression estimates.

Table 4: Depression and Age

	Correlation	OLS Coefficient
Age	0.129	0.050***
		(0.004)
Constant		17.88***
Observations	7,398	7,398
R-squared		0.017

Note: Standard errors in parentheses. *** p<0.01

Age is positively associated with depression ($r = 0.129$, $p < 0.001$). Each additional year of age corresponds to a 0.05-point increase in the Kessler score. While statistically significant, the explanatory power is modest, accounting for only 1.7% of the variation in depression scores.

4.2 Treatment Effects

I now turn to the primary causal analysis: estimating the effect of group therapy on depression using Wave 2 data. Table 5 presents unadjusted summary statistics by treatment status.

Table 5: Depression by Treatment Status (Wave 2)

Group	N	Mean	SD
Control	3,194	17.52	5.59
Treatment	3,191	16.98	5.35
Difference		−0.54	

Raw means indicate that treated individuals have lower depression scores than control individuals. The difference of 0.54 points corresponds to approximately 0.1 standard deviations.

4.2.1 Average Treatment Effect

Table 6 reports regression estimates of the average treatment effect (ATE).

Table 6: Average Treatment Effect of Group Therapy on Depression

	Kessler Score
Treatment	-0.538^{***}
	(0.157)
Constant	17.521^{***}
	(0.114)
Observations	6,385
Clusters	4,352
R-squared	0.002

Note: Robust standard errors clustered at the household level in parentheses. *** p<0.01

The treatment coefficient is -0.54 and statistically significant at the 1% level ($p = 0.001$). This indicates that, on average, individuals in treated households scored 0.54 points lower on the Kessler-10 scale relative to the control group.

While an effect size of 0.1 standard deviations is modest, it represents a meaningful shift in mental health distribution at the population level. The negative sign confirms the hypothesis that GT sessions reduce psychological distress.

4.2.2 Heterogeneity by Gender

A critical policy question is whether treatment effects vary by demographic characteristics. Table 7 reports estimates from the interaction model specified in Equation (2).

Table 7: Treatment Effect Heterogeneity by Gender

	Kessler Score
Treatment (β_1)	-0.129
	(0.206)
Woman (β_2)	0.813***
	(0.168)
Treatment × Woman (β_3)	-0.724***
	(0.235)
Constant (β_0)	17.061***
	(0.147)
Observations	6,385
Clusters	4,352
R-squared	0.005
Estimated Treatment Effects:	
Men: β_1	-0.129 (p = 0.531)
Women: $\beta_1 + \beta_3$	-0.853*** (p < 0.001)

Note: Robust standard errors clustered at the household level in parentheses. *** p<0.01

The results reveal striking heterogeneity by gender. The coefficient on the interaction term ($\beta_3 = -0.724$) is negative and statistically significant ($p = 0.002$), indicating that the beneficial effect of treatment is substantially larger for women than for men.

The coefficients can be interpreted as follows:

- **Constant ($\beta_0 = 17.06$):** The mean Kessler score for untreated men.

- **Treatment ($\beta_1 = -0.13$):** The treatment effect for men. This estimate is small and statistically indistinguishable from zero ($p = 0.531$), suggesting GT sessions did not significantly reduce depression among men.

- **Woman ($\beta_2 = 0.81$):** The gender gap in the control group; women score 0.81 points higher than men in the absence of treatment.

- **Interaction ($\beta_3 = -0.72$):** The differential treatment effect. Treatment reduces depression by an additional 0.72 points for women relative to men.

The total treatment effect for women, computed as $\beta_1 + \beta_3 = -0.853$, is substantial and highly significant ($p < 0.001$). Women in treated households scored nearly one full point lower on the K10 scale compared to their counterparts in the control group.

5 Discussion and Conclusion

This paper evaluated the effectiveness of a group therapy intervention in rural Ghana using a randomized controlled trial. The analysis yields several key insights.

First, GT sessions were effective at reducing depression in the aggregate. Treated individuals scored 0.54 points lower on the Kessler-10 scale compared to control individuals, a result that is robust to clustering at the household level. This contributes to the evidence base that community-based mental health interventions can be effective in low-resource settings.

Second, and crucially, treatment effects were driven almost entirely by women. Women experienced a significant 0.85-point reduction in depression scores, whereas men showed no statistically significant improvement. The interaction term confirms that this gender difference is statistically robust.

Several mechanisms may explain this heterogeneity. Women may be more receptive to group-based therapeutic approaches that emphasize social connection and shared experience.

Cultural norms in rural Ghana may create higher barriers for men to engage in emotional disclosure within a group setting. Alternatively, the specific content of the GT sessions may have been better aligned with the stressors and coping strategies prevalent among women.

Third, baseline analysis confirms that women report higher levels of depression than men, a finding consistent with global epidemiological data. The intersection of higher baseline need and stronger treatment response suggests that women are a highly efficient target population for such interventions.

Finally, household wealth showed no significant association with depression. This null finding is notable given the theoretical link between poverty and mental health, suggesting that in this specific context, non-material factors—such as social standing, physical health, or family dynamics—may play a more decisive role in psychological well-being.

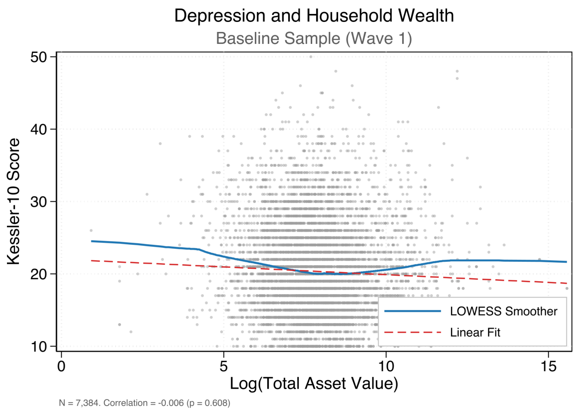
These findings carry clear policy implications. Mental health programs in similar contexts may achieve greater efficacy by specifically targeting women or by adapting curricula to better engage men. While the intervention successfully reduced depression among women, the lack of impact on men highlights the need for alternative approaches—perhaps individual counseling or interventions integrated into economic activities—to reach this demographic. Future research should prioritize unpacking the mechanisms driving this gender divergence to design more inclusive mental health systems.

References

Patel, V., Araya, R., Chatterjee, S., Chisholm, D., Cohen, A., De Silva, M., ... & van Ommeren, M. (2007). Treatment and prevention of mental disorders in low-income and middle-income countries. *The Lancet*, 370(9591), 991–1005.

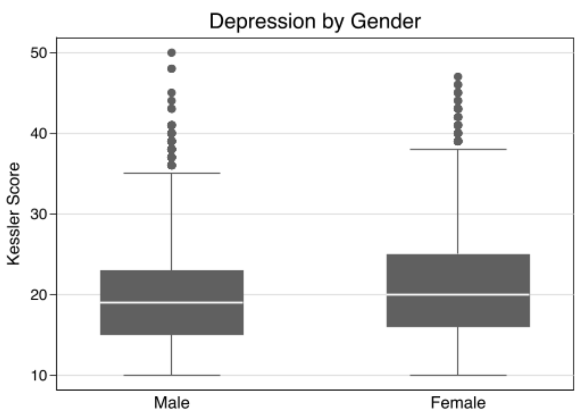
World Health Organization. (2017). *Depression and other common mental disorders: Global health estimates*. Geneva: World Health Organization.

Appendix: Figures

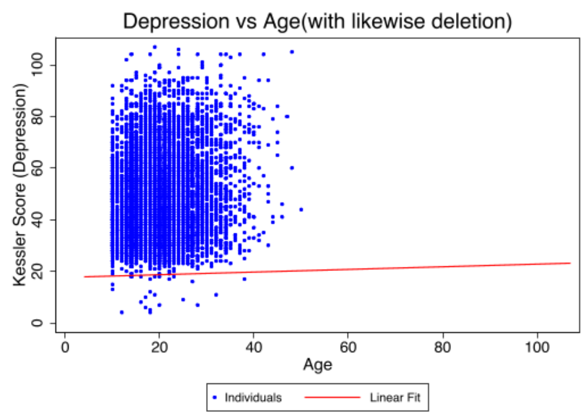


(a) Depression vs. Household Wealth

(b) Depression by Wealth Quartiles



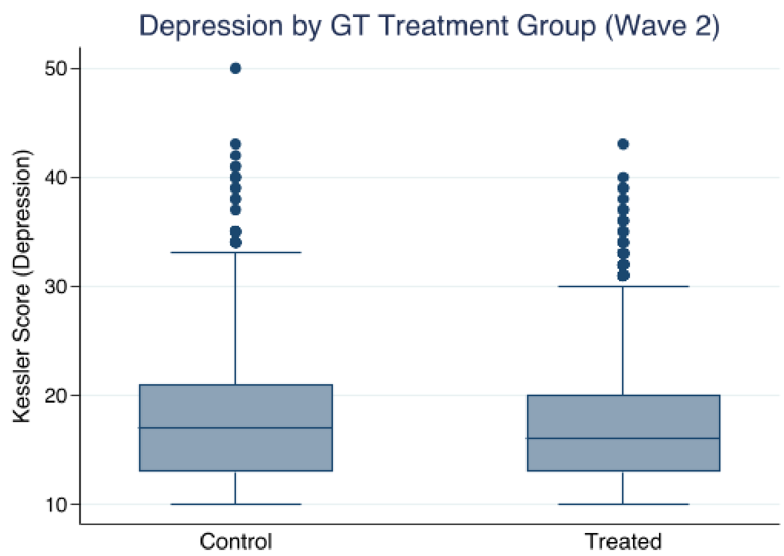
(c) Depression by Gender



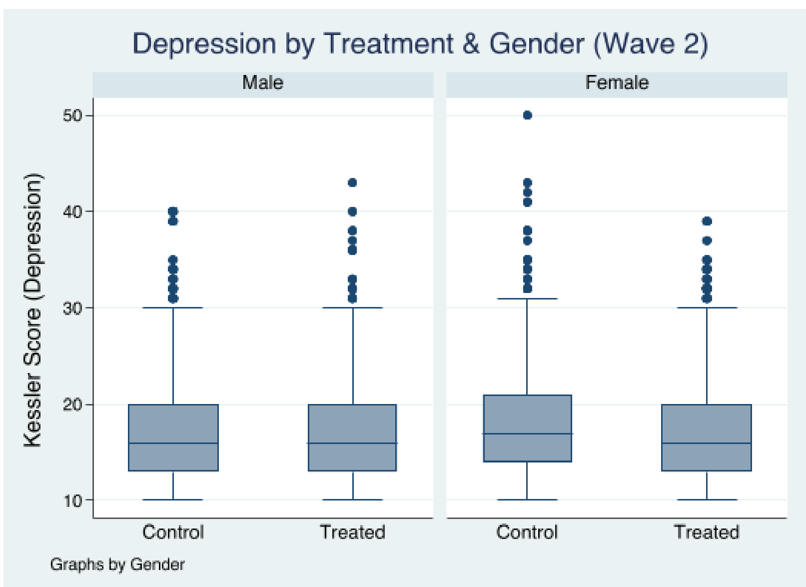
(d) Depression vs. Age

Figure 1: Exploratory Analysis: Depression and Baseline Characteristics (Wave 1)

Notes: Data from Wave 1 (Baseline). (a) Scatter plot of Kessler-10 scores against the natural log of total household assets (GHS). The blue line represents a LOWESS smoother, and the red dashed line represents a linear fit. (b) Box plot of Kessler-10 scores across quartiles of total household assets. The box represents the interquartile range (IQR), the internal line is the median, and the whiskers extend to 1.5 $\times$ IQR. (c) Box plot of Kessler-10 scores by gender (0=Male, 1=Female). (d) Scatter plot of Kessler-10 scores against respondent age in years, including a linear fit line. Sample size $N = 7,398$.



(a) Depression by Treatment Status



(b) Depression by Treatment Status and Gender

Figure 2: Treatment Effects on Depression (Wave 2)

Notes: Data from Wave 2 (Endline). (a) Box plot of Kessler-10 scores comparing the Control group versus the Treatment group (Households assigned to Group Therapy). (b) Box plot of Kessler-10 scores separated by treatment status and gender (0=Male, 1=Female). The lower scores in the "Treatment - Female" group illustrate the heterogeneous treatment effect identified in the regression analysis. Outliers are represented by dots beyond the whiskers.